Program 8: Maven to Gradle Migration

Step 1: Create a Maven Project

```
162 directories, 75 files
harith@harith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/MyMavenToGradle
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  2.156 s
[INFO] Finished at: 2026-03-25T09:00:33+05:30
[INFO] ----------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops$
```

Step 2: Edit pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example1</groupId>
<artifactId>MyMavenToGradle</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenToGradle</name>
<url>http://maven.apache.org</url>
<!--  Properties: Customize Java version or plugin versions  -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!--  Build: Configuring plugins and build settings  -->
<build>
<plugins>
<!--  Example: Maven Compiler Plugin to compile Java code  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
```

```
<configuration>
<source>1.6</source>
<target>1.6</target>
</configuration>
</plugin>
<!--  Example: Maven Surefire Plugin to run tests  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example1.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 3: Build and compile the project

mvn clean install

```
harith@harith-VirtualBox:~/devops/MyMavenToGradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example1:MyMavenToGradle >-------------------
[INFO] Building MyMavenToGradle 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenToGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/MyMavenToGradle/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/harith/devops/MyMavenToGradle/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/harith/devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenToGradle ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenToGradle ---
[INFO] Building jar: /home/harith/devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenToGradle ---
[INFO] Installing /home/harith/devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar to /home/harith/.m2/repository/com/example1/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.0-
[INFO] Installing /home/harith/devops/MyMavenToGradle/pom.xml to /home/harith/.m2/repository/com/example1/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.778 s
[INFO] Finished at: 2026-03-25T09:07:52+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/MyMavenToGradle$
```

Step 4: Execute: gradle init –type pom.xml

this gives an error as xml format is not accepted by gradle

```
harith@harith-VirtualBox:~/devops/MyMavenToGradle$ gradle init --type pom.xml

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported types: 'basic', 'groovy-application', 'groovy-library', 'java-application', 'java-library', 'pom', 'scala-libra

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log output. Run with --scan to get full insights.

* Get more help at https://help.gradle.org

BUILD FAILED in 1s
2 actionable tasks: 2 executed
harith@harith-VirtualBox:~/devops/MyMavenToGradle$ ls
gradle  gradlew  gradlew.bat  pom.xml  src  target
```

Step 5: Execute: gradle init –type pom

```
harith@harith-VirtualBox:~/devops/MyMavenToGradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 4s
2 actionable tasks: 1 executed, 1 up-to-date
harith@harith-VirtualBox:~/devops/MyMavenToGradle$
```

Step 8: Run the Application

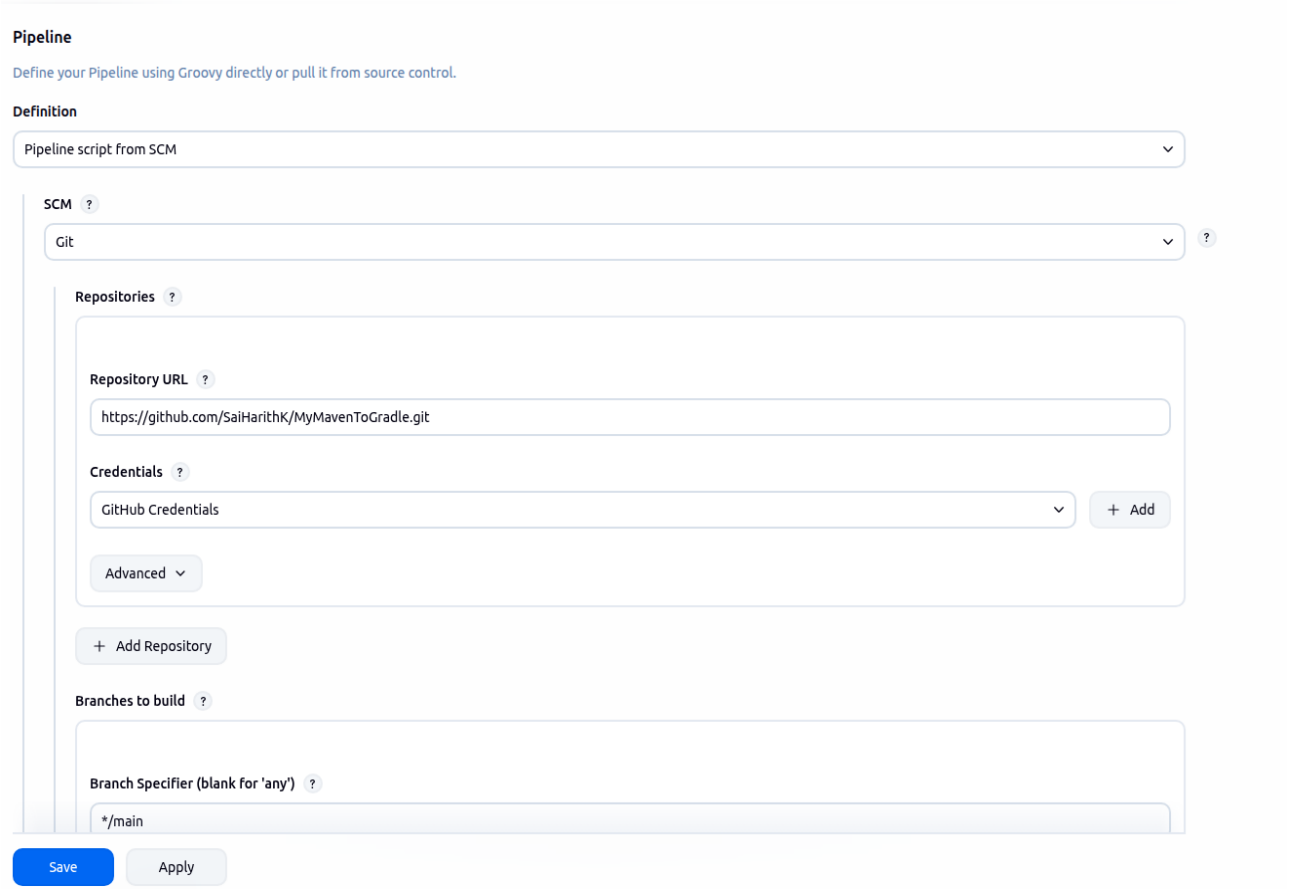
Run the application using: mvn exec:java

```
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$ mvn exec:java
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenFirefoxApp >-------------------
[INFO] Building MyMavenFirefoxApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenFirefoxApp ---
```

Step 9: Pipeline

-login to jenkins account

-create a new item and configure the following settings



Step 10: Edit Jenkins file

-Create a Jenkinsfile and add the following code and push it to github repo

```
pipeline {
   agent any
   tools {
      maven 'Maven'
   }
  stages {
      stage('Checkout') {
         steps {
            git branch: 'main', url: 'https://github.com/SaiHarithK/MyMavenToGradle.git'
         }
      }

      stage('Build') {
         steps {
```

```
            sh 'mvn clean package'
        }
    }
    stage('Test') {
        steps {
            sh 'mvn test'
        }
    }
    stage('Check Target Folder') {
        steps {
            sh 'ls -l target'
        }
    }
    stage('Run Application') {
        steps {
            sh 'java -jar target/MyMavenToGradle-1.0-SNAPSHOT.jar'
        }
    }
  }
  post {
    success {
        echo 'Build and execution successful!'
    }
    failure {
        echo 'Build failed!'
    }
  }
}
```

Step 11: Build

Click on build now and check if its successfull

mavenToGradle

Ad

Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Check Target Folder	Run Application	Declarative: Post Actions
Average stage times: (full run time: ~23s)	1s	191ms	1s	9s	2s	399ms	396ms	156ms
#2 Apr 13 20:19 1 commit	1s	187ms	1s	10s	5s	677ms	677ms	186ms
#1 Apr 13 20:18 No Changes	1s	196ms	1s	8s failed	136ms failed	122ms failed	115ms failed	127ms